Mixed Numbers in Real Life
Answer Key

Grade 4–5

Quick Answers

1. $3 + 1 \cdot \frac{1}{2}$ mi
2. $2 + 1 \cdot \frac{1}{4}$ ft
3. $4 + 1 \cdot \frac{1}{3}$ cups
4. $2 + \frac{2}{3}$ L
5. $4 + \frac{2}{5}$ kg
6. $2 + 1 \cdot \frac{1}{2}$ m
7. $2 + 1 \cdot \frac{1}{2}$ kg
8. $3 + \frac{3}{4}$ hours
9. $2 + \frac{2}{3}$ m
10. $1 + 1 \cdot \frac{1}{2}$ cups

Curriculum

Level: Grade 4–5
4.NF.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 9, 10*
5.NF.A.1 – *problem 8*
Difficulty 1–5 of 5 across 10 tagged checks.

1. Priya hikes $1\frac{1}{4}$ mi and then $2\frac{1}{4}$ mi. Total?
Solution: Add wholes and parts separately: wholes $1 + 2 = 3$; fourths $\frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$. Total: $3 + \frac{1}{2}$, so she hikes $\boxed{3\frac{1}{2}\text{ mi}}$

2. A $4\frac{3}{8}$ ft board loses a $2\frac{1}{8}$ ft piece. How much is left?
Solution: Subtract wholes and parts separately: wholes $4-2 = 2$; eighths $\frac{3}{8}-\frac{1}{8} = \frac{2}{8} = \frac{1}{4}$. Board left: $\boxed{2\frac{1}{4}\text{ ft}}$

3. Recipes use $2\frac{2}{3}$ cups and $1\frac{2}{3}$ cups of flour. Total?
Solution: Wholes: $2+1 = 3$. Thirds: $\frac{2}{3} + \frac{2}{3} = \frac{4}{3} = 1\frac{1}{3}$ — more than a whole, so regroup it into the whole-number part: $3 + 1\frac{1}{3} = 4\frac{1}{3}$. Flour needed: $\boxed{4\frac{1}{3}\text{ cups}}$

4. A pitcher holds $3\frac{5}{6}$ L; guests drink $1\frac{1}{6}$ L. How much is left?
Solution: Wholes: $3 - 1 = 2$. Sixths: $\frac{5}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$. Lemonade left: $\boxed{2\frac{2}{3}\text{ L}}$

5. Watermelons weigh $1\frac{3}{5}$ kg and $2\frac{4}{5}$ kg. Total weight?
Solution: Wholes: $1 + 2 = 3$. Fifths: $\frac{3}{5} + \frac{4}{5} = \frac{7}{5} = 1\frac{2}{5}$ — regroup the extra whole: $3 + 1\frac{2}{5} = 4\frac{2}{5}$. Total: $\boxed{4\frac{2}{5}\text{ kg}}$

6. A spool holds $5\frac{1}{4}$ m; Deshi uses $2\frac{3}{4}$ m. How much rope is left?
Solution: We cannot take $\frac{3}{4}$ from $\frac{1}{4}$, so regroup one whole: $5\frac{1}{4} = 4\frac{5}{4}$. Now subtract: wholes $4 - 2 = 2$; fourths $\frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$. Rope left: $\boxed{2\frac{1}{2}\text{ m}}$

7. Jo's pumpkin weighs $6\frac{1}{8}$ kg, Ben's $3\frac{5}{8}$ kg. How much heavier is Jo's?

Solution: $\frac{5}{8}$ is more than $\frac{1}{8}$, so regroup: $6\frac{1}{8} = 5\frac{9}{8}$. Wholes: $5 - 3 = 2$. Eighths: $\frac{9}{8} - \frac{5}{8} = \frac{4}{8} = \frac{1}{2}$. Difference: $\boxed{2\frac{1}{2}\text{ kg}}$

8. Mira works $1\frac{1}{2}$ hours, then $2\frac{1}{4}$ hours. Total time?

Solution: The parts have different denominators, so rewrite halves as fourths: $\frac{1}{2} = \frac{2}{4}$. Wholes: $1 + 2 = 3$. Fourths: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$. Total time: $\boxed{3\frac{3}{4}\text{ hours}}$

9. Zoe needs $4\frac{1}{3}$ m of fabric and has $1\frac{2}{3}$ m. How much more does she need?

Solution: Missing amount $= 4\frac{1}{3} - 1\frac{2}{3}$. Regroup: $4\frac{1}{3} = 3\frac{4}{3}$. Wholes: $3 - 1 = 2$. Thirds: $\frac{4}{3} - \frac{2}{3} = \frac{2}{3}$. She needs $\boxed{2\frac{2}{3}\text{ m}}$ more.

10. A 6-cup bowl gets $2\frac{3}{4}$ cups of orange juice and $1\frac{3}{4}$ cups of pineapple juice. How much more fills it?

Solution: First the juice so far: $2\frac{3}{4} + 1\frac{3}{4} = 3 + \frac{6}{4} = 3 + 1\frac{2}{4} = 4\frac{1}{2}$ cups. Then the space left: $6 - 4\frac{1}{2} = 5\frac{2}{2} - 4\frac{1}{2}$, which leaves $\boxed{1\frac{1}{2}\text{ cups}}$